Testing for Homogeneity in Panel Data Models with Discrete Unobserved Heterogeneity

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1 Introduction

It has been well known that classic Wilks' Theorem for the asymptotic distribution of LRT statistics cannot be usually applied in finite mixture models, due to the presence of nuisance parameters that are not identified under the null hypothesis and mixing parameters on the boundary. In this paper, we derive the asymptotic distribution of likelihood ratio test (LRT) statistics for testing homogeneity in panel data models with normal mixture under difference in location. We show how reparametrization technique introduced in Kasahara and Shimotsu (2012) (KS12, hereafter) can be used to study its asymptotic behaviours under the framework of Andrews (1999, 2001) (A99 and A01, hereafter) to eliminate these issues.

The goal of this paper is to recite the general results suggested in Andrews (1999, 2001) and KS12 and verify that analyzing the asymptotic distribution of LRT statistic in mixture panel models can be performed in a similar framework. For both completeness of the paper and learning purpose, I have tried to make this paper self-contained as much as possible by completing all the necessary proof. For some of the steps, I directly refer to A99, A01, and KS12 when the procedure is too technical to be recalled.

The paper is organized as follows. In Section 2, I present the model specification, key assumptions, and discuss some important properties of dynamic panel data models in mixture settings. Section 3 introduces several challenges in finding the

solution and identify the main strategy used in this paper to solve them. I also discuss some key papers in literature and their contribution to the methods used in this paper. Since these papers play strong roles in our arguments, I provide a detailed exposition on their methods and strategies. Section 4 derives asymptotic distribution of the LRT statistic. The paper concludes in Section 5, suggesting possible improvement and future directions for further research. All proofs for propositions and lemmas can be found in Appendix.

Throughout the paper, $\to_d$ and $\to_p$ mean "converges in distribution" and "converges in probability" respectively. When the right-hand side of $\to_d$ denotes a probability distribution, it means convergence in distribution to some random variable that has a distribution of the latter. EX for some random variable X denotes an expectation $\int XdP$ (if it is well defined) for some probability measure P that is absolutely continuous with respect to Lebesgue measure μ. $\|\cdot\|$ is a usual Euclidean norm for vectors and its induced matrix norm for matrices. $:=$ means equals by definition. For a scalar-valued function $f(\mathbf{x})$ for a $k \times 1$ vector $\mathbf{x} = (x_1, ..., x_k)'$, $\nabla_{\mathbf{x}} f(\mathbf{x})$ denotes the $k \times 1$ gradient of f with respect to $\mathbf{x}$. $\nabla_{x_i} f(\mathbf{x})$ denotes the partial derivative of $f(\mathbf{x})$ with respect to x_i for $i = 1, ..., n$. $\nabla_{x_i^{(n)}} f(\mathbf{x})$ denotes the nth partial derivative of $f(\mathbf{x})$ with respect to x_i for $i = 1, ..., n$.

2 Model specification and main hypotheses

Suppose we have n individuals, each with T length of observations, where each agent belongs to one of two types of unobserved heterogeneity indexed by $\{1, 2\}$. The probability of being the first type is determined by $\alpha \in \Theta_\alpha \subset [0, 1]$, with the following probability of being the second type as $1 - \alpha$. The observation on period t from ith agent whose latent variable is $x_i \in \{1, 2\}$, y_{it}, is generated by

$$y_{i,t} = \mu_{x_i} + \epsilon_{i,t}$$

where $\epsilon_{i,t}$ is an iid error term generated by $N(0, 1)$, and $\mu_{x_i} \in \Theta_\mu$ for some $\Theta_\mu \subset \mathbb{R}$. Let Θ denote the collection of model specifications in a vector form $\vartheta = (\mu_1, \mu_2, \alpha)$ for

$\mu_1, \mu_2 \in \Theta_\mu$ and $\alpha \in \Theta_\alpha$. Collect the T observations from an ith individual as $\mathbf{y}_i = (y_{i,1}, y_{i,2}, ..., y_{i,T})$, and let $\mathcal{Y}_n = (\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_n)$ denote the collection of observations from all individuals. Then, the likelihood function of the model given $\vartheta \in \Theta$ is

$$L_n(\mathcal{Y}_n, \vartheta) = \prod_{i=1}^{n} L(\mathbf{y}_i, \theta, \alpha)$$

with

$$L(\mathbf{y}, \vartheta) = \alpha f(\mathbf{y}; \mu_1) + (1-\alpha) f(\mathbf{y}; \mu)$$
$$f(\mathbf{y}; \mu) = \prod_{t=1}^{T} \phi(y_{i,t} - \mu)$$

where $\phi(x)$ is the density of the standard normal. Throughout the paper, we shall let $l_n(\mathcal{Y}_n, \vartheta)$ and $l(\mathbf{y}, \vartheta)$ denote the logarithm of $L_n(\mathcal{Y}_n, \vartheta)$ and $L(\mathbf{y}, \vartheta)$ respectively. For notational brevity, we abbreviate $\mathcal{Y}$ and $\mathbf{y}$. We shall call this model as a *mixture panel model* throughout the paper.

The main null hypothesis is that

$$H_0 : \alpha(1-\alpha)(\mu_1 - \mu_2) = 0$$

which can be decomposed into two null hypotheses.

$$H_{01} : \mu_1 - \mu_2 = 0$$
$$H_{02} : \alpha(1-\alpha) = 0$$

so that $H_0 = H_{01} \cup H_{02}$.

3 Literature review and main strategy

3.1 Literature review

The standard Wilk's Theorem cannot be usually directly applied in testing for homogeneity in mixture models due to the fact that under the null hypothesis $H_{02} : \alpha(1-\alpha) = 0$ the true model contains the parameter α on the boundary of the parameter space $\Theta_\alpha := [0, 1]$. Also, the existence of a nuisance parameter,

μ_2, makes the analysis of asymptotic distribution for the LRT statistic even more complicated as μ_2 is not identified under the null hypothesis $\alpha = 1$ in H_{02} (likewise, μ_1 is not identified for the case $\alpha = 0$). Some papers avoid this non-interior and non-identifiability problem by assuming that $\alpha \in [\epsilon, 1-\epsilon]$ for some $\epsilon \in (0, 1/2)$, and proceed to test H_{01} (Qin and Smith, 2004) or assume that mixing parameters μ_1, μ_2 are strictly separated (Ghosh and Sen, 1985). Even if we relax the condition and test the null hypothesis of H_{01} only, non-identifiability issues are still present as α is not identified in H_{01} either, making analysis still difficult. Chen and Chen (2001) prove the general result without the assumption on separation of mixing parameters for the asymptotic distribution of LRT statistic for homogeneity.

A99 and A01 have suggested a new rigorous methodology that can be used to test when a parameter is on the boundary of the maintained hypothesis with the presence of nuisance parameters that are not identified under the null hypothesis. The main strategy used in both papers is to derive an asymptotic distribution of an M-estimator in a shifted and rescaled parameter space by approximation from a certain suitable quadratic expansion of log-likelihood functions. Under some regularity conditions, the MLE of the original model can be characterized by the asymptotic distribution of the M-estimator from the reparametrized parameter space. The results are valid even when the true parameter is on the boundary of the parameter space; the case it is not on a boundary closely resembles a classic argument used to induce local asymptotic normality. Our procedure on log-likelihood expansion and quadratic approximation for testing H_{01} is based on Andrew's suggestion in A99 and A01.

Andrew's fraemwork in A99 and A01 can play a role in testing H_{01} as done in Cho and White (2007). However, under H_{02}, the parameter α is on a boundary with the presence of nuisance parameters μ_1 and μ_2 that are not identified. To see this, note that when the mixing parameter for the component, α, is zero or one, then the model is equivalent regardless of the choice of the parameter for component one or two, respectively. This is exactly why, as mentioned in the first paragraph of this chapter, that previous literature have extensively focused on testing for H_{01} instead.

This is the key argument linking H_{01} and H_{02} together to deduce the asymptotic distribution of LRT for the whole null hypothesis H_0 as a whole, and this problem is to remain as an open question. In this paper, we apply the restriction that $\Theta_\alpha \subset (0,1)$ as done in Qin and Smith (2004) to proceed to test for H_{01} as the main alternative hypothesis.

Also, in order to employ the results from A99 and A01, one needs to handle the degeneracy of Fisher information matrices, which is usually considered as one of the undesired properties in mixture models. We overcome this problem by reparametrization of the original parameter space, which yields a suitable quasi-score and information matrices as done in KS12.

3.2 Main strategy

Using the methodology introduced above, we adopt the following strategy as a main framework to deduce the asymptotic distribution of LRT statistic for testing homogeneity in mixture panel data models.

First, we show consistency of the MLE under mixture panel data models and prove useful results from reparametrized log-likelihood functions that can be used in subsequent sections. Second, based on these results, we construct a likelihood function expansion for a fixed $\alpha \in (0, 1/2)$ up to the fourth order and transform it into a suitable quadartic form under additional rescaling and shifting that forms a convex reformed parameter space Λ. Analytic expressions for each term will be also provided. Third, we partition the reshaped reparametrized parameter in Λ into two parts, one related with a variable of interest and the other for nuisance parameters to verify that the quadratic expansion obtained satisfies sufficient conditions suggested in A99. Fourth, using arguments from A99, we show that the asymptotic distribution of the LRT statistics is that of the minimizer of the quadratic term of the reparametrized quadratic expansion. Lastly, we showing that this minimizer follows that of $[\max(0, N(0,1))]^2$, concluding that the asymptotic distribution of LRT test statistics is $[\max(0, N(0,1))]^2$.

4 Deriving the asymptotic distribution of the LRT statistics

Following the approach from KS12, we first define new parameters

$$\eta := \alpha\mu_1 + (1-\alpha)\mu_2$$
$$\lambda := \mu_1 - \mu_2$$

and denote the space of these new parameters as $\Theta_\eta \subset \mathbb{R}$ and $\Theta_\lambda \subset \mathbb{R}$. We shall often collect η and λ as a vector ψ such that $\psi = (\eta, \lambda)$. Note that once $\alpha > 0$ is known, the full parameters can be recovered from η by

$$\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \begin{pmatrix} \eta + (1-\alpha)\lambda \\ \eta - \alpha\lambda \end{pmatrix}$$

We now study the behaviours of the LRT statistic under H_{01}. Let $\psi^* = (\eta^*, \lambda^*) = (\eta^*, 0)$ with $\mu_1^* = \mu_2^* := \mu^* \in \Theta_\mu$ denote the true parameter under this null hypothesis. Fix $\alpha \in (0,1)$ and perform Taylor expansion $l_n(\psi, \alpha)$ around (ψ^*, α) up to the four times yields

$$\begin{aligned} l_n(\psi,\alpha) - l_n(\psi^*,\alpha) = {} & \nabla_\eta l_n(\psi^*,\alpha)(\eta-\eta^*) + \frac{1}{2}\nabla_{\lambda^{(2)}} l_n(\psi^*,\alpha)\lambda^2 + \\ & \frac{3}{3!}\nabla_{\eta\lambda^{(2)}} l_n(\psi^*,\alpha)\lambda^2(\eta-\eta^*) + \frac{1}{2}\nabla_{\eta^{(2)}} l_n(\psi^*,\alpha)(\eta-\eta^*)^2 + \\ & \frac{1}{4!}\nabla_{\lambda^{(4)}} l_n(\psi^*,\alpha)\lambda^4 + R_n(\psi,\alpha) \end{aligned} \tag{1}$$

where $R_n(\psi, \alpha)$ is a remainder term whose analytic form and property is given in Proposition 2 and its proof. Now, we normalize the terms in Equation 1 so that it has an appropriate quadratic structure to apply the methodology A99 has adopted. This is done by shifting and rescaling the parameter space. First, define a new parameter

$$t_n(\psi,\alpha) = \begin{pmatrix} \sqrt{n}(\eta-\eta^*) \\ \sqrt{n}\alpha(1-\alpha)\lambda^2 \end{pmatrix} \tag{2}$$

which induces the following shifted and rescaled parameter space for (η, λ^2)

$$\Lambda_n := \sqrt{n}(\Theta_\eta - \eta^*) \times \sqrt{n}\alpha(1-\alpha)\Theta_\lambda^2 \tag{3}$$

for each $n \in \mathbb{N}$ and fixed η^* and α, where $\Theta_\lambda^2 := \{\lambda^2; \lambda \in \Theta_\lambda\}$. For notational brevity, we shall abbreviate parameters in $t_n(\psi, \alpha)$ and denote it as t_n. Let $\widetilde{\nabla}_{t_n} l(\mathbf{y}_i, \psi, \alpha)$ denote the corresponding quasi-score of t_n as

$$\widetilde{\nabla}_{t_n} l(\mathbf{y}_i, \psi, \alpha) := \begin{pmatrix} \nabla_\eta l(\mathbf{y}_i, \psi, \alpha) \\ \frac{1}{2}\nabla_{\lambda^{(2)}} l(\mathbf{y}_i, \psi, \alpha)/(\alpha(1-\alpha)) \end{pmatrix} \tag{4}$$

with

$$G_n = n^{-1/2} \sum_{i=1}^{n} \widetilde{\nabla}_{t_n} l(\mathbf{y}_i, \psi^*, \alpha)$$

Define the corresponding information matrix for t_n by

$$\mathcal{I}_n = \begin{pmatrix} \mathcal{I}_{n\eta} & \mathcal{I}_{n\eta\lambda} \\ \mathcal{I}_{n\eta\lambda} & \mathcal{I}_{n\lambda} \end{pmatrix}$$

where

$$\begin{aligned} \mathcal{I}_{n\eta} &= -n^{-1}\nabla_{\eta^{(2)}} l_n(\psi^*, \alpha) \\ \mathcal{I}_{n\eta\lambda} &= -\frac{1}{2} n^{-1} \frac{\nabla_{\eta\lambda^{(2)}} l_n(\psi^*, \alpha)}{\alpha(1-\alpha)} \\ \mathcal{I}_{n\lambda} &= -\frac{2}{4!} n^{-1} \frac{\nabla_{\lambda^{(4)}} l_n(\psi^*, \alpha)}{\alpha^2(1-\alpha)^2} \end{aligned}$$

Using these expressions, the fourth order expansion in Equation 1 can be converted to the following quadratic form as desired after some algebra:

$$l_n(\psi, \alpha) - l_n(\psi^*, \alpha) = t_n' G_n - \frac{1}{2} t_n' \mathcal{I}_n t_n + R_n(\psi, \alpha) \tag{5}$$

Before summarizng the properties of this quadratic form, we shall first prove some useful properties of the mixture panel model.

Lemma 1. *The mixture panel model satisfies the following properties:*
(a) A log-likelihood function $l_n(\mu, \alpha)$ is smooth and is continuously differentiable up to four times almost surely in μ.
(b) The model is identifiable; $f(\mathbf{y}; \mu_1) = f(\mathbf{y}; \mu_2)$ implies $\mu_1 = \mu_2$.
(c) For $\alpha \in (0, 1)$, $E|\nabla^k f(\mathbf{x}; \psi^*, \alpha)/f(\mathbf{x}; \psi^*, \alpha)| < \infty$ for $k = 1, 2, 3, 4$.

With the claims above, the following regularity assumption yields consistency result:

Assumption 1. 1. Θ_μ is compact.

2. μ^* is in the interior of Θ_μ.

3. $\Theta_\alpha = [\epsilon_1, 1-\epsilon_1]$ for some $\epsilon_1 \in (0, 1/2)$.

Remark 1. The first two assumptions in Assumption 1 are standard. Note that the last part of Assumption 1 ensures to test the null hypothesis under restriction that $\Theta_\alpha \subset (0,1)$ as done in Qin and Smith (2004). This yields the alternative form of a null hypothesis $H_0 : \mu_1 - \mu_2 = 0$.

Proposition 1. *Let Θ_0 be the collection of parameters $\vartheta^* = (\mu_1, \mu_2, \alpha) \in \Theta_\mu \times \Theta_\mu \times \Theta_\alpha$ under the null hypothesis. Under Assumption 1, MLEs are consistent. That is, $\hat{\vartheta}_n \to_p \vartheta^*$.*

Now we can fully characterize the properties of the reparametrized quadratic expansion as follows, with some useful results for calculation:

Lemma 2. *The reparametrized density satisfies the followings:*

(a) $\nabla_\lambda f(\mathbf{x}; \psi^*, \alpha) = 0, \qquad \nabla_\lambda l(\mathbf{x}; \psi^*, \alpha) = 0, \qquad \nabla_{\eta\lambda} \ln f(\mathbf{x}; \psi^*, \alpha) = 0$

(b) $E[\nabla_{\lambda^{(2)}} \ln f(\mathbf{x}; \psi^*, \alpha)] = 0, E[\nabla_{\lambda^{(3)}} f(\mathbf{x}; \psi^*, \alpha)] = 0, E[\nabla_{\eta\lambda^{(2)}} f(\mathbf{x}; \psi^*, \alpha)] = 0$

(c) $E[\nabla_{\lambda^{(4)}} \ln f(\mathbf{x}; \psi^*, \alpha)] = -3E[(\nabla_{\lambda^{(2)}} \ln f(\mathbf{x}; \psi^*, \alpha))^2]$

Proposition 2. *Suppose Assumption 1 holds. For a fixed $\alpha \in (0,1)$, the mixture panel model satisfies the followings:*
(a) Equation 1 holds. Furthermore, for any $\delta > 0$, $\limsup_{n\to\infty} P(\sup_{\psi\in\Theta_\psi, \|\psi-\psi^*\|\le\epsilon} |R_n(\psi,\alpha)| > \delta(1 + \|t_n(\psi,\alpha)\|)^2) \to 0$ as $\epsilon \to 0$.
(b) $\mathcal{I}_n \to_p \mathcal{I}$, where $\mathcal{I}$ is finite and positive definite.
(c) $G_n \to_d N(0, \mathcal{I})$.

Remark 2. (i) Rather than showing stochastic boundedness of the remainder term, we prove a stronger result in Proposition 2 (a) to satisfy the sufficient condition Assumption 2* suggested in A99.
(ii) It has been well-known that normal mixture models with a structural variance

suffers from nondegenency in a Fisher information matrix. Even after reparametrization procedure as in this paper, the problem is persistent. However, it can be actually shown that $\mathcal{I}$ is still finite and positive definite with the presence of common variance in mixture panel models under our reparametrization technique. Due to the space constraint, our model tests for homogeneity only in mean.

4.1 Asymptotic distribution of LRT statistics for testing H_{01}

In this subsection, we follow the procedure suggested in A99 and obtain the asymptotic distribution of the LRT statistic after shifting and rescaling the parameter space given in (3). Note that the true value of $(\lambda^*)^2 = 0$ is on the boundary of the parameter space $\Theta_\lambda^2 = \mathbb{R}_+$ under H_{01}; this is where A99 provides a key to the problem. First, normalizing expressions in Equation 5 by defining $Z_n = \mathcal{I}_n^{-1} G_n$, we have

$$l_n(\psi, \alpha) - l_n(\psi^*, \alpha) = \frac{1}{2} Z_n' \mathcal{I}_n Z_n - \frac{1}{2}[t_n(\psi, \alpha) - Z_n]' \mathcal{I}_n [t_n(\psi, \alpha) - Z_n] + R_n(\psi, \alpha) \quad (6)$$

The main argument of A99 is that finding the supremum of the above expression is equivalent to equivalent to finding the infimum of the quadratic term whose solution $\hat{t} \in \Lambda := \mathbb{R} \times \Theta_\lambda^2$ can be approximated by some shifted and rescaled parameter space $\Lambda_n/\sqrt{n}$ (in this case, (3)). That is, we have

$$\begin{aligned} \sup_{\psi \in \Theta_\psi} 2[l_n(\psi, \alpha) - l_n(\psi^*, \alpha)] &= Z_n' \mathcal{I}_n Z_n - \inf_{t \in \Lambda} (t - Z_n)' \mathcal{I}_n (t - Z_n) + o_p(1) \\ &\to_d Z' \mathcal{I} Z - \inf_{t \in \Lambda} (t - Z)' \mathcal{I} (t - Z) = \hat{t}' \mathcal{I} \hat{t} \end{aligned} \quad (7)$$

for some random variable $\hat{t}$, where

$$\Lambda = \mathbb{R} \times \Theta_\lambda^2 \quad (8)$$

It turns out that the term relevant to λ in t determines the whole behaviour of $\hat{t}$ as found in the subsequent Proposition 3.

To achieve this, we partition Z and G as follows as in A99:

$$Z = \begin{pmatrix} Z_\eta \\ Z_\lambda \end{pmatrix}, \qquad G = \begin{pmatrix} G_\eta \\ G_\lambda \end{pmatrix} \quad (9)$$

with corresponding variance terms

$$\mathcal{I}^*_\eta := E(G^2_\eta), \qquad \mathcal{I}^*_{\lambda\eta} := E(G_\lambda G_\eta), \qquad \mathcal{I}^*_{\eta\lambda} := \mathcal{I}^*_{\lambda\eta}, \qquad \mathcal{I}^*_\lambda := E(G^2_\lambda)$$

Note that $Z_\lambda = G_{\lambda.\eta}/\mathcal{I}^*_{\lambda.\eta}$, where

$$\begin{aligned} G_{\lambda.\eta} &:= G_\lambda - \mathcal{I}^*_{\lambda\eta} G_\eta / \mathcal{I}^*_\eta \\ \mathcal{I}^*_{\lambda.\epsilon} &:= G_\lambda - \mathcal{I}^{*2}_{\lambda\eta}/\mathcal{I}^*_\eta = \mathrm{Var}(G_{\eta.\lambda}) = (\mathrm{Var}(Z_\lambda))^{-1} \end{aligned} \tag{10}$$

so that similar to (7) we define t_λ as

$$g_\lambda(\hat{t}_\lambda) = \inf_{t_\lambda \in \Lambda_\lambda} g_\lambda(t_\lambda) \tag{11}$$

for

$$g_\lambda(t_\lambda) := (t_\lambda - Z_\lambda)^2 \mathcal{I}_{\lambda.\eta}$$

where $\Lambda_\lambda = \Theta^2_\lambda$. Now we are ready to present the main result for this subsection. Let $\hat{\psi}_{n,\alpha} = (\hat{\eta}, \hat{\lambda})$ denote the MLE for a given α, i.e.,

$$\hat{\psi}_{n,\alpha} := \arg \sup_{\psi \in \Theta_\eta \times \Theta_\lambda} l_n(\psi, \alpha)$$

Given $\epsilon_1 \in (0, 1/2)$, define the likelihood ratio test statistic for H_{01} as

$$LR_{n,01}(\epsilon) = \sup_{\alpha \in [\epsilon, 1-\epsilon]} 2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_{n,0}(\hat{\mu}_0)] \tag{12}$$

where $l_{n,0}(\mu_0) = \sum_{i=1}^n \log f(\mathbf{y}_i, \mu_0)$ is the one-component log-likelihood function with its corresponding MLE $\hat{\mu}_0$.

Now we have the main results:

Proposition 3. *Suppose that Assumption 1 holds. Then, for each $\alpha \in (0, 1)$, we have the followings:*

(a) $2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_n(\psi^*, \alpha)] \to_d \hat{t}^2_\lambda \mathcal{I}^*_{\lambda.\eta} + G^2_\eta / \mathcal{I}^*_\eta$ *where $\hat{t}_\lambda$ is defined as in (11).*

(b) $2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_{n,0}(\hat{\mu}_0)] \to_d \hat{t}^2_\lambda \mathcal{I}^*_{\lambda.\eta}$ *where $\hat{t}^2_\lambda \mathcal{I}^*_{\lambda.\eta}$ follows* $[\max[N(0,1), 0]]^2$.

Remark 3. (a) The decomposition in (10) arises from decomposing the LRT statistic into two parts, one for $l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_n(\psi^*, \alpha)$ and the other for $l_{n,0}(\hat{\mu}_0) - l_n(\psi^*, \alpha)$.

(b) Since the rescaled space is generated by λ^2, which is a convex function, the cone Λ is convex. This condition might be violated if the space of nuisance parameters is multi-dimensional.

(c) Note that Part (c) implies that the asymptotic distribution of the LRT statistic under H_{01} is invariant under the choice of α. This yields the following corollary, which is our main theorem:

Corollary 1. *Suppose that Assumption 1 holds. Then,* $LR_{01}(\epsilon_1) \to_d [\max[N(0,1),0]]^2$

5 Conclusion

We have shown that under the strong assumption $\alpha \in [\epsilon_1, 1-\epsilon_1]$, the LRT statistic for homogeneity in mixture panel models asymptotically follows $[\max[N(0,1),0]]^2$, for a given $\epsilon_1 \in (0, 1/2)$. Note that in the toy example studied in this paper, only the mean is different across components. The reason why I chose panel data models as a main model is that it actually contains a number of attractive characteristics that are both interesting and worthwhile to study for statistical analysis. I conclude this paper by outlining problems that I have not been able to solve in this paper that are also possble improvements that I plan to investigate further so that (hopefully) I can answer the following problems in the next year gradually:

(1) Presence of structural parameters. Usual panel data models have additional structural parameters that account for individual differences. In mixture models, these parameters are going to be nuisance parameters that have to be dealt with as well. A01 formulates the asymptotic distribution of LRT statistics with the presence of nuisance structural parameters and will work as a main reference. The procedure is essentially similar.

(2) Presence of unknown varying/common variance. We have made a strong heteroskedastic assumption in our model. The analysis can be generalized for a known arbitrary variance $\sigma^2 > 0$ after rescaling, as done in Qin and Smith (2004). If variance is unknown, it becomes a nuisance parameter to be estimated. If variance

is not different across components, the variance becomes a structural parameter. A notable characteristic of the panel data models is that the Fisher information matrix is not singular, unlike in normal mixture models. When variance is different across components, we need extra assumptions. We have to add additional assumptions then, such as compactness of the parameter space or different estimation method using penalty functions.

(3) Dynamic panel data models. The strength of panel data models lies in its ability to estimate time-varying behaviours of individuals. It would be interesting to set the autoregressive coefficients switching across components. I expect that the analysis would require a similar procedure as done in analyzing tests for homogeneity in regime switching models, which can be also studied under the framework this paper has adopted as seen in Qu and Zhuo (2016).

(4) VAR models and testing for arbitrary finite number of components. Testing for homogeneity in mixtures would be even more complicating in multivariate cases. This corresponds to the case where our panel data models allow correlation between observations within an agent, or each observation is multidimensional. Testing for the arbitrary number of components with this framework has been done in regression context in Kasahara and Shimotsu (2015).

A Proofs for Theorems

A.1 Proof of Proposition 1

We show consistency by considering the MLE for $l_n(\vartheta)$, $\hat{\vartheta}_n$ as an M-estimator for

$$M_n(\vartheta) = n^{-1}\sum_{i=1}^{n}\log[\alpha f(\mathbf{y}_i;\mu_1) + (1-\alpha)f(\mathbf{y}_i;\mu_2)]$$

with its limit $M(\vartheta) := E\log[f(\mathbf{y}_i;\mu^*)]$ accordingly, which is finite as for every $\mu \in \Theta_\mu$

$$E\log[f(\mathbf{y}_i;\mu^*)] = E\left[\frac{T}{2}(2\pi) - \frac{1}{2}\sum_{t=1}^{T}(\mathbf{y}_{i,t}-\mu^*)^2\right] < \infty \tag{13}$$

due to finiteness of $E\mathbf{y}_{i,t}$ and $E\mathbf{y}_{i,t}^2$ by the model assumption that $\mathbf{y}_{i,t} \sim N(0, \mu^*)$ under the null hypothesis. By Assumption 1 (a), Θ_μ is compact, and $\log f(\mathbf{x}; \mu)$ is continuous in μ from Lemma 1 (a). Also, $|\log f(\mathbf{x}, \mu)|$ is bounded above by $\log f(\mathbf{x}, \hat{\mu})$ for some $\hat{\mu}$ uniformly, which exists and is integrable by (13), by compactness of the parameter space and continuity of the log-likelihood function. Thus, by the uniform law of large numbers (Newey and McFadden (1989), Lemma 2.4), we have

$$\sup_{\vartheta \in \Theta} \|M_n(\theta) - M(\theta)\| \to_p 0$$

Also, note that $M(\theta)$ is bounded above by $M(\theta^*)$ for all $\vartheta \in \Theta$ as the Kullback-divergence is zero only at the true parameter. Classic consistency results from M-estimator theory (Theorem 5.7 in Van der Vaart, 2000) with identifiability condition in Lemma 1 (b) concludes the consistency claim.

A.2 Proof of Proposition 2

Taylor expandability for Equation 1 follows from Lemma 1 (a), and with the help from Lemma 2 it can be asily seen that Equation 1 can be deduced with the remainder term that is of stochastic order $O_p(n)$ from WLLN. Showing the sufficient condition for A99 is technical, so we leave the details to the original paper KS12 directly. The only unclear part in their proof is to show that

$$\sup_{\psi \in \Theta_\psi \cap \mathcal{N}} \left| n^{-1} \nabla^{(4)} l_n(\psi, \alpha) - E[\nabla^{(4)} \ln f(\mathbf{x}_i; \psi, \alpha)] \right| = o_p(1)$$

and

$$E\nabla^{(4)} \ln f(\mathbf{x}_i; \psi, \alpha) \text{ is continuous in } \psi \in \Theta_\psi \cap \mathcal{N}$$

for a neighbourhood $\mathcal{N}$ of ψ^*. The proof for the first claim is similar to the one used for consistency in Proposition 1; it suffices to show that ULLN holds. $\nabla^{(4)} l_n(\psi, \alpha)$ is continuous in ψ from the fact that the density is smooth and the probability of having density to be zero is zero. Also, $|\nabla^{(4)} l_n(\psi, \alpha)|$ is bounded above by $|\nabla^{(4)} l_n(\hat{\psi}, \alpha)|$ for some $\hat{\psi}$ uniformly, which exists and is integrable, by compactness of the closure of the parameter space and continuity of the log-likelihood function, for all $\psi \in$

$\overline{\Theta_\psi \cap \mathcal{N}}$ (the space of continuous functions in a compact space is also compact.). The compactness of the closure of $\Theta_\psi \cap \mathcal{N}$ follows by the fact that Θ_ψ is bounded. Thus, by the uniform law of large numbers (Lemma 2.4 in Newey and McFadden, 1989), the claim follows. The second claim follows from the dominated convergence theorem, where the existence of a dominating function is given in the proof for the first claim.

Convergence result of part (b) is immediate from the law of large numbers, by defining

$$\mathcal{I} = \begin{pmatrix} \mathcal{I}_\eta & \mathcal{I}_{\eta\lambda} \\ \mathcal{I}_{\eta\lambda} & \mathcal{I}_\lambda \end{pmatrix} \tag{14}$$

where

$$\begin{aligned} \mathcal{I}_\eta &= -E\nabla_{\eta^{(2)}} l_n(\psi^*, \alpha) \\ \mathcal{I}_{\eta\lambda} &= -\frac{1}{2} E \frac{\nabla_{\eta\lambda^{(2)}} l_n(\psi^*, \alpha)}{\alpha(1-\alpha)} \\ \mathcal{I}_\lambda &= -\frac{2}{4!} E \frac{\nabla_{\lambda^{(4)}} l_n(\psi^*, \alpha)}{\alpha^2(1-\alpha)^2} \end{aligned}$$

as the expectations are finite by Lemma 1 (a) and the fact that the parameter space is compact (hence the reparametrized parameter space is bounded.) from Assumption 1. We now argue that the resulting matrix $\mathcal{I}$ is invertible. Compare the expression given in (4) and (14). By Bartlett's identity, we have

$$E[\widetilde{\nabla} l(\mathbf{y}_i, \psi^*, \alpha) \widetilde{\nabla} l(\mathbf{y}_i, \psi^*, \alpha)'] = -\mathcal{I} \tag{15}$$

Note that the identity in the last term, $\mathcal{I}_\lambda = \frac{1}{2} E(\nabla_{\lambda^{(2)}} l(\mathbf{y}_i, \psi, \alpha)/(\alpha(1-\alpha)))^2$ can be also seen from Lemma 2 (c) by cancelling the multiplication by three.

From (15), it suffices to check non-collinearity of the elements in (4), which can be found by direct calculation. The formula for $\widetilde{\nabla} l(\mathbf{y}_i, \psi^*, \alpha)$ in (4) can be alternatively represented as

$$\begin{pmatrix} \nabla_\eta l(\mathbf{y}_i, \psi, \alpha) \\ \frac{1}{2} \nabla_{\lambda^{(2)}} l(\mathbf{y}_i, \psi, \alpha)/(\alpha(1-\alpha)) \end{pmatrix} = \begin{pmatrix} \dfrac{\nabla_\mu f(\mathbf{x}; \phi^*, \alpha)}{f(\mathbf{x}; \phi^*, \alpha)} \\ \dfrac{1}{2} \dfrac{\nabla_{\mu^{(2)}} f(\mathbf{x}; \phi^*, \alpha)}{f(\mathbf{x}; \phi^*, \alpha)} \end{pmatrix}$$

so from the expressions $f_\mu(\mathbf{x}; \mu), f_{\mu^{(2)}}(\mathbf{x}; \mu)$ that are given explicitly in the proof of

Lemma 1 and the fact that $\mathbf{x}_t$ is a random varaible that follows the standard normal with μ, we have non-singularity condition of $\mathcal{I}$.

Note that $E\nabla_{\lambda^2} l(\mathbf{y}_i, \psi^*, \alpha) = E\nabla_\eta l(\mathbf{y}_i, \psi^*, \alpha) = 0$. Hence, Part (c) follows from the multivariate CLT by Proposition 2 (b).

A.3 Proof of Proposition 3

Part (a) is a direct verification of Corollary 1(c) of A99 comparing the results we have by far. The conditions of Corollary 1(c) are Assumption 2-5, 7, and 8 in A99 by considering $(B_T, Dl_T(\theta_0), \mathcal{J}_T, Z_T, (H\mathcal{J}_*^{-1}H')^{-1})$ in A99 as $(\sqrt{n}, \sum_{i=1}^n \widehat{\nabla}_{t_n} l(\mathbf{y}_i, \psi^*, \alpha), \mathcal{I}_n, Z_n, \mathcal{I}_{\lambda.\eta})$ in our paper. To distinguish the assumptions given in A99 from ours, we put a letter A in front of the assumptions given in A99. For brevity, we do not recite all the conditions explicitly unless a new definition is introduced.

A-Assumption 2 is satisfied with the sufficient condition A-Assumption 2*, Proposition 3 (a). A-Assumption 3, which states that $\mathcal{I}_n \to_d \mathcal{I}$ for non-singular $\mathcal{I}$ and $Z_n \to_d Z$ for some well-defined Z and follows from Proposition 2 (b) and (c).

A-Assumption 4, $t_n = O_p(1)$ has to be proven explicitly. First, by consistency and the definition of the MLE from Proposition 1 and the fact that the log-likelihood functions are continuous from Lemma 1, we have that

$$o_p(1) \leq l_n(\hat{\psi}, \alpha) - l_n(\psi^*, \alpha)$$

Using the second-order expression form we have in (5) and from Proposition 2 (a), we have

$$\begin{aligned} o_p(1) &\leq l_n(\hat{\psi}, \alpha) - l_n(\psi^*, \alpha) \\ &= t_n' G_n - \frac{1}{2} t_n' \mathcal{I}_n t_n + (1 + \|t_n\|)^2 o_p(1) \end{aligned}$$

Rewrite this by normalizing t_n by $\tilde{t}_n := \mathcal{I}_n^{1/2} t_n$. We have then

$$o_p(1) \leq \tilde{t}_n' \mathcal{I}_n^{1/2} Z_n - \frac{1}{2}\|\tilde{t}_n\|^2 + (1 + \|\mathcal{I}_n^{-1/2}\tilde{t}_n\|)^2 o_p(1)$$

Recall that $Z_n = \mathcal{I}_n^{-1} G_n$. We first argue that $\mathcal{I}_n^{1/2} Z_n = \mathcal{I}_n^{-1/2} G_n = O_p(1)$ by the repeated use of law of large numbers. The limit of $\mathcal{I}_n$, $\mathcal{I}$ has no zero-element. To

see this, recall the proof of Proposition 2(b); if it had contained a zero element then it would not have been invertible by construction. Thus, we have

$$E\nabla_{\eta^{(2)}} l_n(\psi^*, \alpha), E\frac{\nabla_{\eta\lambda^{(2)}} l_n(\psi^*, \alpha)}{\alpha(1-\alpha)}, E\frac{\nabla_{\lambda^{(4)}} l_n(\psi^*, \alpha)}{\alpha^2(1-\alpha)^2} \neq 0$$

so $\mathcal{I}_n^{-1/2} G_n = (\sqrt{n}\mathcal{I}_n^{-1/2})(n^{-1/2}G_n)$ is $O_p(1)$ by applying WLLN on elements in $(\sqrt{n}\mathcal{I}_n^{-1/2})$ and $n^{-1/2}G_n$ then recall Slutsky's lemma. Using this information, we have the following inequality:

$$\begin{aligned} o_p(1) &\leq O_p(\|\tilde{t}_n\|) - \frac{1}{2}\|\tilde{t}_n\|^2 + o_p(\|\tilde{t}_n\|) + o_p(\|\tilde{t}_n^2\|) + o_p(1) \\ &= \|\tilde{t}_n\| O_p(1) - \frac{1}{2}\|\tilde{t}_n\|^2 + o_p(\|\tilde{t}_n\|) + o_p(\|\tilde{t}_n^2\|) + o_p(1) \end{aligned}$$

Rearranging this inequality yields

$$\|\tilde{t}_n\|^2 \leq 2\|\tilde{t}_n\| O_p(1) + o_p(1)$$

this implies that $\|\tilde{t}_n\| = O_p(1)$. To see this, first we write the $O_p(1)$ term in RHS as ξ. Then we have

$$\|\tilde{t}_n\|^2 - 2\|\tilde{t}_n\|\xi + \xi^2 \leq \xi^2 + o_p(1)$$

which can be factored into

$$(\|\tilde{t}_n\| - \xi)^2 \leq \xi^2 + o_p(1)$$

Taking the square root on both sides and recalling that $\xi = O_p(1)$ verifies the claim that $\|\tilde{t}_n\| = O_p(1)$. Now, recall that $\tilde{t}_n = \mathcal{I}_n^{1/2} t_n$. Proposition 2 (b) yields the conclusion from this claim.

To verify A-Assumption 5, or its sufficient condition A-Assumption 5*, we need to show that the space of Λ_n/n is locally equal to Λ. According to A99, a set $\Gamma \subset \mathbb{R}^s$ is locally equal to a set $\Lambda \subset \mathbb{R}^s$ if $\Gamma \cap C(0, \epsilon) = \Lambda \cap C(0, \epsilon)$ for some $\epsilon > 0$ where $C(0, \epsilon)$ is a cube centred at 0 whose side is 2ϵ. This can be easily seen by recalling the definition of Λ_n and Λ from (3) and (8):

$$\Lambda_n = (\Theta_\eta - \eta^*) \times \alpha(1-\alpha)\Theta_\lambda^2$$

$$\Lambda = \mathbb{R} \times \Theta_\lambda^2$$

A-Assumption 7 and A-Assumption 8 for block diagonality of $\mathcal{I}$ and $\Lambda_1 = \mathbb{R}$ follow from Proposition 2 (c) and our construction of Λ. This establishes part (a) from A99.

For part (b) the first claim that $2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_{n,0}(\hat{\mu}_0)] \to_d \hat{t}_\lambda^2 \mathcal{I}^*_{\lambda.\eta}$ can be found by decomposing the LRT statistic as the sum of two LRT statistics follows:

$$\begin{aligned} 2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_{n,0}(\hat{\mu}_0)] &= 2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_n(\psi^*, \alpha)] - 2[l_{n,0}(\hat{\mu}_0) - l_n(\psi^*, \alpha)] \\ &= 2[l_n(\hat{\psi}_{n,\alpha}, \alpha) - l_n(\psi^*, \alpha)] - 2[l_{n,0}(\hat{\mu}_0) - l_{n,0}(\mu^*)] \end{aligned}$$

now it remains to show that $2[l_{n,0}(\hat{\mu}_0) - l_{n,0}(\mu^*)]$ asymptotically follows the distribution of that of $G_\eta^2/\mathcal{I}^*_\eta$. The proof is repetitive of what we have done so far; taking the fourth order expansion as in (1) under $\lambda = 0$ reduces the expression to the second order. Perform reparametrization as done in (2). The second element is not needed as it reduces to the second order expression, so t_n becomes scalar. This results in absence of the second term (6), leaving only the first term, which is equivalent to $G_\eta^2/\mathcal{I}^*_\eta$. It can be also seen as the special case where only the last term $G_\eta^2/\mathcal{I}^*_\eta$ in Proposition 3 (b) is left due to the fact that λ term is gone.

The second claim that $\hat{t}_\lambda^2 \mathcal{I}^*_{\lambda.\eta}$ follows $[\max[N(0,1), 0]]^2$ can be shown as follows: first, note that $\hat{t}_\lambda$ is the minimizer of

$$(t_\lambda - Z_\lambda)^2 \mathcal{I}^*_{\lambda.\eta} = (t_\lambda - Z_\lambda)^2/(Var(Z_\lambda))$$

from (11). There are two possibilities: $Z_\lambda \geq 0$ and $Z_\lambda < 0$. When $Z_\lambda \geq 0$, $t_\lambda = Z_\lambda$ minimizes the term by making it zero. When $Z_\lambda < 0$, as $t_\lambda \geq 0$ since $t_\lambda \in \Lambda = \{\lambda^2; \lambda \in \Theta_\lambda\}$, the unique solution is $t_\lambda = 0$. Hence,

$$\hat{t}_\lambda = Z_\lambda \mathbb{1}(Z_\lambda \geq 0)$$

holds. Therefore, $\hat{t}_\lambda^2 \mathcal{I}^*_{\lambda.\eta}$ follows $[\max[N(0,1), 0]]^2$.

B Proofs for Lemmas

B.1 Proof of Lemma 1

Part (a) follows from continuity and smoothness of exponential functions, and the fact that $f(\mathbf{x}_t, \mu) = 0$ happens with zero probability for all μ by the properties of Lebesgue measure.

For part (b), we first suppose that $f(\mathbf{x}; \mu_1) = f(\mathbf{x}; \mu_2)$ for some μ_1, μ_2. Taking the logarithm on both sides yields

$$-2\sum_{t=1}^{T}(\mu_1 - \mu_2)\mathbf{x}_t + T(\mu_1^2 - \mu_2^2) = 0$$

which holds when $\mu_1 = \mu_2$. Suppose for contradiction that $\mu_1 \neq \mu_2$. Dividing both sides by $\mu_1 - \mu_2$ yields

$$-2\sum_{t=1}^{T}\mathbf{x}_t + T(\mu_1 + \mu_2) = 0$$

which holds iff

$$\sum_{t=1}^{T}\mathbf{x}_t = \frac{1}{2}T(\mu_1 + \mu_2)$$

Since $(\mathbf{x}_t)_{t=1}^T$ is an iid collection of normal random variables, their sum is a normal random variable. Hence, this occurs with zero probability by the property of Lebesgue measure, implying that $\mu_1 = \mu_2$ must be the case. Thus, the model is identifiable.

Part (c) follows from the direct computation by differentiating the density by the original parameter:

$$\begin{aligned}
f_\mu(\mathbf{x}; \mu) &= \sum_{t=1}^{T}(\mathbf{x}_t - \mu)f(\mathbf{x}; \mu) \\
f_{\mu^{(2)}}(\mathbf{x}; \mu) &= \sum_{t=1}^{T}\mathbf{x}_t f_\mu(\mathbf{x}; \mu) - Tf(\mathbf{x}; \mu) - T\mu f_\mu(\mathbf{x}; \mu) \\
f_{\mu^{(3)}}(\mathbf{x}; \mu) &= \sum_{t=1}^{T}\mathbf{x}_t f_{\mu^{(2)}}(\mathbf{x}; \mu) - Tf_\mu(\mathbf{x}; \mu) - Tf_\mu(\mathbf{x}; \mu) - Tf_{\mu^{(2)}}(\mathbf{x}; \mu)
\end{aligned}$$

substituting expressions $f_\mu(\mathbf{x}; \mu)$ and $f_{\mu^{(2)}}(\mathbf{x}; \mu)$ by the original $f(\mathbf{x}; \mu)$ in above formula and the fact that $(\mathbf{x}_t)_{t=1}^T$ follows standard normal with iid and thus $\sum_{t=1}^T \mathbf{x}_t +$

k for arbitrary constant $k \in \mathbb{R}$ also follows normal (so that it has finite first moment) imply the result. The case for $k = 4$ follows from the similar argument recursively.

B.2 Proof of Lemma 2

The results from the following algebra: we first show part (a). First, by the chain rule using our definition that $\lambda = \mu_1 - \mu_2$, we have that

$$\nabla_\lambda f(\mathbf{x}; \psi^*, \alpha) = [(1-\alpha)\alpha \nabla_\mu f(\mathbf{x}; \mu^*) - \alpha(1-\alpha)\nabla_\mu f(\mathbf{x}; \mu^*)] = 0$$

under the null hypothesis that the true mean values are $\mu_1^* = \mu_2^* = \mu^*$. Hence, by the chain rule we also have

$$\nabla_\lambda l(\mathbf{x}; \psi^*, \alpha) = 0$$

This also yields the expression for $\nabla_{\eta\lambda} \ln f(\mathbf{x}; \psi^*, \alpha)$:

$$\begin{aligned}\nabla_{\eta\lambda} \ln f(\mathbf{x}; \psi^*, \alpha) &= \nabla_\lambda \frac{\nabla_\mu f(\mathbf{x}; \mu)}{f(\mathbf{x}; \mu)}\bigg|_{\mu=\mu^*} \\ &= \frac{[\nabla_\lambda \nabla_\mu f(\mathbf{x}; \mu)] f(\mathbf{x}; \mu) - \nabla_\mu f(\mathbf{x}; \mu) \nabla_\lambda f(\mathbf{x}; \psi^*, \alpha)}{f^2(\mathbf{x}; \mu)}\bigg|_{\mu=\mu^*} = 0\end{aligned}$$

the first term is zero as a similar reasoning for $\nabla_\lambda f(\mathbf{x}; \psi^*, \alpha)$, and the last term is zero by the fact that $\nabla_\lambda f(\mathbf{x}; \psi^*, \alpha) = 0$ from the last line. This concludes part (a).

For part (b), note that

$$E[\nabla_\lambda l(\mathbf{x}; \psi, \alpha)] = 0$$

by Lemma 1 (a). Since $\nabla_\lambda l(\mathbf{x}; \psi, \alpha) f(\mathbf{x}; \psi, \alpha)$ is continuously differentiable from the same lemma, the change of order in differentiation is possible. Hence, differentiating both sides by λ, we have

$$\int [\nabla_{\lambda^{(2)}} l(\mathbf{x}; \psi, \alpha) + \nabla_\lambda l(\mathbf{x}; \psi' \alpha) \nabla_\lambda l(\mathbf{x}; \psi, \alpha)] f(\mathbf{x}; \psi, \alpha) d\mu = 0$$

Evaluating this expression at ψ^* gives $E[\nabla_{\lambda^{(2)}} \ln f(\mathbf{x}; \psi^*, \alpha)] = 0$, evaluating at ψ^* after differentiating with respect to λ gives $E[\nabla_{\lambda^{(3)}} f(\mathbf{x}; \psi^*, \alpha)] = 0$, and evaluating at ψ^* after differentiating with respect to λ and then η gives $E[\nabla_{\eta\lambda^{(2)}} f(\mathbf{x}; \psi^*, \alpha)] = 0$. This concludes part (b).

Part (c) follows from evaluating this expression at ψ^* after differentitating with respect to λ twice.

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